Madhani Rohith

ee23btech11038@iith.ac.in | rohithmadhani@gmail.com | +91 7780113182 | LinkedIn | GitHub

EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.05
FIITJEE Junior College	Class XII (Telangana State Board of Intermediate Education)	2023	95.50%
FIITJEE School	Class X (Board of Secondary Eduation, Telangana)	2021	10.00%

WORK EXPERIENCE

Technology Analyst Intern | *Morgan Stanley, Bangalore*

- Built config-driven Python/Airflow pipelines to migrate 26 tables (50-50,000 rows) from IBM DB2 to Snowflake, enabling business users to build analytics dashboards on Power BI directly from Snowflake
- Designed daily-scheduled DAGs to incrementally sync fresh data as it landed in DB2, completing full pipeline runs in under a minute versus an estimated multi-hour manual migration process
- Leveraged Snowflake's cloud data warehouse architecture (separation of storage and compute) to enable fast, scalable querying for downstream Power BI dashboards without impacting source DB2 performance
- Used an existing config-driven framework to onboard new tables without writing custom pipeline code per table, reducing migration effort and pipeline duplication across the dataset

PROJECTS

Order Book and ITCH 5.0 Market-Data Pipeline | *Personal Project*

- Built a C++20 limit order book (price-indexed levels, intrusive FIFO order queues, pre-allocated object pool, open-addressing order-ID hash map), fed by a zero-copy NASDAQ ITCH 5.0 parser over a lock-free SPSC queue. Replayed a 20M-message synthetic feed (about 14M book updates), with the final book state matching an independent generator's expected values.
- Benchmarked on pinned cores using rdtscp timing with timer overhead subtracted: book updates at **p50 177 ns and p99 579 ns**, with the pipeline sustaining **3.5M updates/s**. At a paced 1M msgs/s, queue-plus-apply latency was **p50 411 ns** (WSL2, i7-12650H).
- Replaced std::unordered_map with a custom flat hash map for order lookup, cutting p50 latency by **37%** (283 -> 177 ns), p99 by **37%** (925 -> 579 ns) and p99.9 by **73%** (3.1 µs -> 0.84 µs). All figures are medians of 3 runs on an identical replay.

Portfolio Optimization with Cumulative Prospect Theory Utility via Convex Optimization

Course Project, Convex Optimization

- Implemented 3 optimization algorithms (MM, CCP, Projected Gradient Ascent) to solve a non-concave CPT portfolio allocation problem, using a convex-minus-convex decomposition to build tractable concave minorants
- Extended the baseline with an Adam-based optimizer, achieving the highest CPT utility (0.0475 vs. 0.0403 for MM/CCP) at competitive runtime
- Built a scaling benchmark on 6,000 synthetic returns, showing GPU-parallelized GA/Adam converge where classical convex solvers fail
- Validated results via rolling backtest on real market data, with GA/Adam outperforming mean-variance and convex-relaxation baselines

Exploring Adaptive Noise Detection for Zero-Shot Test-Time Adaptation

Course Project, Computer Vision

- Conducted a systematic ablation study of AdaND, a state-of-the-art Zero-Shot Noisy Test-Time Adaptation (ZS-NTTA) framework, dissecting 4 core design dimensions (thresholding, detector architecture, training signal,

inference-time decisions) across 11 method variants on CLIP-based vision-language models
- Designed a confidence-margin training scheme that excludes ambiguous pseudo-labeled samples near the decision threshold, yielding the strongest gain (+0.69% avg AccH, +4.27% on the hardest CIFAR-100+Places pair) by directly reducing pseudo-label noise rather than reweighting or softening labels
- Diagnosed failure modes in 8 alternative design choices (e.g., GMM thresholding, MLP detectors, uncertainty deferral) through controlled experiments, identifying pseudo-label quality from a frozen CLIP encoder as the fundamental performance ceiling of the framework

Selective Noise Canceller | *Course Project, EE2800*

- Implemented an adaptive noise cancellation system in MATLAB using the Recursive Least Squares (RLS) algorithm, dynamically updating filter coefficients to track non-stationary noise characteristics in real time
- Designed an IIR filter for partial noise suppression and combined it with the adaptive RLS stage to maximize output SNR gain
- Benchmarked adaptive vs. fixed filtering approaches, demonstrating [40 dB] SNR improvement over the baseline IIR-only method

SKILLS

Languages: C/C++, Python, MySQL, VS Code, Git/Github

Libraries: Numpy, Pandas, Tensorflow, PyTorch, cvxpy, OpenCV

RELEVANT COURSES

Computer Science: Intro to Programming, Database Management Systems, Data Structures and Applications

Mathematics: Calculus, Differential equations, Vector Calculus, Probability and Random Variables, Complex Variables, Convex Optimization, Statistical Inference, Information Theory

AI/ML: Intro to AI, Matrix Theory, Pattern Recognition and Machine Learning, Deep Learning, Computer Vision, Natural Language Processing